# David J. Castillo-Cornejo Ph.D.

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## EDUCATION

**Universidad Nacional Autónoma de México (UNAM)** **Mexico City, Mexico**
**Instituto de Fisiología Celular** *April 2009*
*Biomedical Sciences Doctoral Program; Molecular Microbiology and Biophysics*

**Universidad Nacional Autónoma de México (UNAM)** **Mexico City, Mexico**
**Faculty of Sciences, Department of Biology.** *April, 2003*

*Bachelor of Sciences, Microbiology*

## WORK EXPERIENCE

**Arizona State University** **Tempe Arizona, USA**
*Research Scientist, Biodesign Institute. Center of Mechanims of Evolution.* *May 2023 – Jul 2024*

I withheld multiple roles as a Research Scientist including training and supervising undergrad and graduate students, postdocs and other members to the lab in learning and implementing molecular microbiology techniques for genetic assays, development of mutants in *E. coli* for flagellar stators motors re-moderation studies. I generated fluorescent protein fusions for sub cellular localization and implemented and optimized Genetically Encoded Voltages Indicators (GEVIs) for membrane potential measurements in *E. coli* cells using confocal fluorescent microscopy. I created several protein tag fusions for cloning, over expressing and purification of membrane proteins channels for their reconstitution in artificial liposomes for further biophysical measurements and characterization.

**Internet of Production Alliance** **USA-Mexico (Remote)**
*Panel Reviewer, "Open know where Awards"* *Mar 2023 – Jun 2023*

I selected and supervised the prospective projects that are involved in mapping the digital manufacturing capabilities across Africa and LATAM teams bringing the best proposals to become eligible for the Award.

**Pontífica Universidad Católica de Chile.** **CDMX,-Santiago de Chile, (Remote)**
*Academic Project Administrator,* *Mar 2022-May2022*
*Instituto de Ingeniería Biológica y Médica*

I **c**oordinated the activities related to the contents and documentation of a Chan-Zuckerberg funded project: LIBRE_hub (Latin American Hub for Bioimaging through Open Hardware) Provided contact for speakers and audiences. Coordinated the organization, meetings and the repository for allocating reports, web site prototypes, and generated demands for online talks and workshops.

**Lab. Molecular Genetics, Epigenetics, Development and Evolution of Plants, Institute of Ecology/Institute of Physics- UNAM / Center for Complexity Sciences (C3), UNAM** **Mexico City , Mexico** *May-Dec 2017*
*Postdoctoral researcher*

I developed a Robotic system for single-cell genetic expression profiling through histological preparation and Laser Microdissection Microscopy, retrieval of nucleic acids in developing roots. I constructed and developed an automated system for the histological inclusion of vegetal tissues for structural and functional genomic studies. The prototype consisted of a microcontroller-operated (arduino/raspberry pi) heated and motorized system for washing, fixation, and inclusion of delicate tissues. The device was built using additive construction technologies (3D printing) and consists of a volume graduated syringe pump, a couple of peristaltic pumps, a PID temperature-controlled heater, and a flow meter. Once included the tissues were collected for microtome dissection, staining, and retrieval of total RNA using laser microdissection (Nikon eclipse). Total recovered RNA was submitted to specific cDNA RT-PCR amplification using real-time PCR for later quantization of transcription-targeted genes.

**Customized Premium Products S.A. de C.V.** **Mexico City , Mexico**
*Business Unit Manager* *Oct. 2014-April 2015*

I directed a team of specialists in public health: Economists (1) Biochemists (2) Medical Doctors (1) and Epidemiologists (2), in conjunction with the sales team (Key Account Managers; 6), operations managers (2), and Market analysts (3). We offered consulting services for Market Managers, Product Managers, medical directors, and other stakeholders in the pharmaceutical industry by providing strategic market intelligence solutions through an epidemiological investigation based on clinical evidence, a systematic review of scientific literature, quantitative epidemiological modeling of several chronic degenerative diseases and other indications of highly specialized medical treatments.

-We offered consulting services for Market Managers, Product Managers, medical directors, and other stakeholders in the pharmaceutical industry providing strategic market intelligence solutions through an epidemiological investigation based on clinical evidence, a systematic review of scientific literature, quantitative epidemiological modeling of several chronic degenerative diseases and other indications of highly specialized medical treatments.
- We delivered a market profile of new drugs and/or incoming medical or commercial pipelines for local and regional markets (Mexico and LATAM), including different treatments for cancer, reproductive medicine, diabetes, cardiovascular diseases, depression, HIV, medical devices (POC devices), dressings. etc.
-We retrieved information through panels of medical specialists (Delphi method) and primary sources of the Mexican healthcare system and other countries in LATAM.
.-We Provided dynamic analysis of Prescription adherence to Treatments and Economic impact survey of medical treatments;
-We were involved in the elaboration of technical and economic dossiers for the registration submission of new drugs to the
COFEPRIS (Federal Commission for the Protection against Sanitary Risk).
- I Identified opportunity areas, resolved of conflicts, and maintained adherence to pharmacovigilance practices for each project.
-I provided Continuous feedback on the development of projects in weekly or monthly meetings with other business teams CPP.
- I maintained continuous communication with customers and ensure responsiveness to changes and immediate evaluation.

**Projects under my supervision:**

**CLIENTS/ PROJECTS**
**Eli Lilly Ltd,** (Mexico - Colombia) Patients Flow; Market sizing.
**Bristol Myers Squibb/ Boehringer** (México - Colombia) Prescriptions behavior, Product cycle.
**Bayer** (Mexico, Colombia, Chile, Argentina & Venezuela) Patients Flow; Market sizing.
**Roche** (Mexico) Patients Flow; Regulatory Affairs
**Novartis / SEDESA** (Health Secretariat) Patients Flow; Regulatory Affairs, Policy mapping.
**Teva Pharmaceuticals** (Mexico) Regulatory Affairs
**Amgen** (Mexico-USA) Patients Flow; Market sizing.
**Eisai Ltd**. (Mexico-Japan) Patients Flow; Market sizing.

**Achievements during my tenure as DMA manager :**
-I assisted in the recruitment of new staff and reshaping of the Business Unit by the reallocation of roles, adopting a novel scheme to renovate the performance and productivity by implementing new organization roles.
- Implemented an automated platform for project management, CRM, personnel performance, customer service; Data storage; Scheduling; Sales tracking with real-time assessments.
- Trained the operations staff, sales force, directors, and managers of commercial areas for the adoption and implementation of the automated process engineering and project management of each operating unit with real-time prorating.
- Trained and Assisted Salesforce and Pollsters teams with reliable updated medical and scientific information.
- Implemented sales presentations for various services and products, updating and adopting a new corporate image.
- Supervised and trained a new analyst for epidemiological modeling, and a new specialized Key Account manager to deploy sales for the government
sector (Market Access), Generic Drugs, and Medical Devices. (implementation of NOM-177-SSA1-1998)
-Developed a Kaizen management model for tracking processes, identifying bottlenecks, challenges and implemented continuous improvements and fostering communication between production areas.
-Delivered epidemiological models to our clients, with continuous updates.

**Waseda University** **Tokyo, Japan**
*Junior researcher, Department of Physics (Kinosita Lab)* *Nov 2012 – Mar 2014*

Research Project: "Systems biology and biophysics of microbial ATP synthase in artificial liposomes using microfluidic chips";(Research). Microfabrication of surfaces for microfluidics and biosensors using FIB microscopy and PDMS masking; Real-time, high-speed **nanophotometric** measurements of FoF1-ATPase rotation. I developed a Research Project: "Systems biology and biophysics of microbial ATP synthase in artificial liposomes" using microfluidic chips and nano-fabrication of surfaces for microfluidics and biosensors using FIB microscopy and PDMS masking; I overexpressed and purified proteins using diverse chromatography techniques, the resulting protein was used for fluorometry assay, and the synthesis of liposomes and designed the microfluidic chips. I used FRET and fluorescent microscopy for real-time, high-speed nanophotometric measurements of FoF1-ATPase rotation. I prepared and functionalized using chemical conjugation the survey of glass slides, glass gold, and iron/fluorescent **nanoparticles** for attaching large ATPase complexes into chemical and biologically functionalized surfaces (metal-glass) **nanobeads.** I trained Postdoctoral research and graduate students in immunological localization, molecular biology (protein engineering), and biophysical measurements of molecular **nanomotors.**

**Osaka University** **Osaka, Japan**
*Postdoctoral Researcher,* *Jul 2010 – Sep 2012*
*Graduate School of Frontier Biosciences, Osaka University*
*(K. Namba Lab)*

Research project: "Nano high-speed photometry nano bacterial flagellum motor"; (Research and Education).
I developed a research project "Measuring the rotation of the motor of the bacterial flagellum by high-speed **nanophotometry**" a Theoretical and practical study on engine dynamics of the bacterial flagellum and optical tools used for measuring the speed and mechanical parameters I performed molecular biology experiments for protein engineering, western blot localization, fluorescent, TIRF, **nanophotometry** microscopy of gold and plastic **nanobeads** using high seed cameras for the biophysical model integration of rotation of flagellar motors. I trained undergrad students in molecular biology techniques, protein engineering and structural analysis of tertiary structures of protein crystals or related studied transmembranal proton channels.

**Max Planck Institute for Terrestrial Microbiology** **Marbug, Germany**
*Postdoctoral Researcher,* *May 2009 – Jun 2010*

I developed the research project: "Biosynthesis of Type 4 pili assembly of *Myxococcus xanthus*". For that project, I used cellular fractionation using low and high-speed analytical centrifugation for the separation of membranous sub-cellular components. I studied the biosynthetic pathway of molecular motors in *M. Xanthus* TIV pili, I used FRET microscopy for red fluorescent fusion protein probes for intracellular spatial localization. I performed Western-blot on different samples for fractioned membranes. I overexpressed and purified proteins for antibody production using manual and automated IMAC. I generated mutants of the Pili systems and relates regulators or its motor.

## MEDIA AND SCIENCE OUTREACH

**Radio** **Mexico City, Mexico**
*Presidential Radio Station*. (Mexican Federal Government *Sep-Nov 2006*
Broadcast Network for Public Education).

*Invited speaker* to weekly Richard Dorfsmann's Radio Show 'Hear from Mexico' Spanish and English editions,
Topic: "What is Nanotechnology?"

**Press** **Mexico City, Mexico**
Co-founder and Editor in Chief. *Feb 2000-Jul 2001*
*Faculty of* Sciences - UNAM

"El Mecate" ("The String"). Non-profit Publication for the communication of Sciences, arts, and literature. I managed the general direction of the publication contents with a group of corresponding assigned editors: Science in Art and design; Style and proofreading; Sales & Advertising; 'News & Interviews'; 'Arts and Music'; 'Sports'; 'Society &Technology' (under my direct contribution); Jobs & Careers' and Website team. I led and coordinated the meetings with each section manager, editorial board, guest collaborators, and individual columnists. Made reviews of writing & designs for final layout printing; proofread final manuscripts and advertising design. Personal contributions in Dossier articles:"Multi(medios) electrónicos, la colectividad atormentada / Electronic multi(media) a tortured collectivity" (Vols. 1 & 2) July August 2000.

## CIVIL, GOVERNMENT & COMMUNITY PARTICIPATION

**Independent Research** **Mexico City, Phoenix Mexico-USA**
Glyxon Biolabs/ Microchip Technologies *Jul 2023- to date*
*Founder.*

How carbon sequestration is modulated by Mycorrhizal and soil bacteria symbionts in Abies trees? Project funded by experiment.com foundation and with the collaboration of multiple academic, industrial and non academic researchers and volunteers: I coordinate a multidisciplinary team of engineers, ecophysiology researchers and microbiologists in the development of a low-cost environmental data-logger that measures soil respiration and multiple environmental parameters associated to the colonization of mycorrhizal communities in different environments.

**Inter-American Development Bank /Biocurious (Maria Chavez and Rafael Anta), and Glyxon Biolabs.** **Mexico City, (Remote)-USA**
*External Analyst* *Apr. 2022- Nov 2022*

**Inter-American Development Bank /Biocurious** (Maria Chavez and Rafael Anta)
Survey and interviews of the project: "Understanding the challenges of decentralized citizen science Biolabs across Latin America"

**REDES A.C.** **Mexico City, Mexico**
*External Analyst* *Oct 2017- Jan 2017*

Analyst for the project " Assessment of Plagues in Urban Forests of Mexico City" (REDES A.C., SEDEMA, PAOT, and independent civil participants); Designed and proposed the aerial layout for the survey of urban forest patches in Mexico City, using UAVs and multispectral cameras. I designed a simple automated algorithm for remote The ID of specific spectral signatures corresponding to epiphytic plagues in trees; Developed portable micro controller operated devices for *in situ* measurements of CO2, temperature, and soil humidity of urban parks. (Implementation of NOM-059-SEMARNAT-2010)

## ENTREPRENEURSHIP

**Mexico City, Mexico**
*2019- to date*

**GLYXON Biolabs DIY Biospace for biomaterials, Founder, Advisor ([www.glyxon.com](www.glyxon.com))**

Glyxon Biolabs studies micro-algae, plants, and bacterial models to explore further strategies of atmospheric CO2 sequestration and biopolymers biosynthesis by combining biotechnological approaches and computer-based simulations of ecological dynamics of planktonic ecosystems and bacterial consortia, in order to develop novel ways to synthesize scalable and sustainable precursors of biofuels, bioplastics and other derivatives required in the electronics, food, medical and pharmaceutical industries. Our interests include traditional and digital microfluidics; 3D printing cell cultures; biopolymers biosynthetic pathways; simulation of genetic chassis circuits; synthetic biology; CRISPR gene editing and protein engineering in bacterial models and automated plants phenotypification; Lab automation using microcontrollers and tailored lab automation circuits.

Current developments: -'Lab on a chip' Flexible low-cost bioreactor and biosensors for bacteria and micro algae growth monitoring using DIYmicrocontrollers
-EWoD micropump for sequential microlitre volume culture manipulation for isolation and retrieval of byproducts from microbial consortia.
-Optogenetics: *Light-induced biosynthesis based on pDusk/pDawn systems coupled with RFP reporters for the production of low cost molecular biology reagents required in COVID19 detection. ([www.glyxon.com/covid19](www.glyxon.com/covid19)) In partnership with Ottawa BioScience (Ottawa Canada) and funded by JOGL-CRI, Paris, France.*
-*Proteopresso. A simplified automated pressurized recombinant protein purification system (Funded by JOGL, in progress)*
-Tissue Macro Photogrammetry using a customized Motorized Micro Slider for macro focus stacking. In collaboration with Simen E. Sörensen (Applied Procrastination, Oslo, Norway) and Stefan Hermann (CNC Kitchen, Wangen, Germany )
-Bioluminescent reporters of bacterial biosynthetic pathways using *Vibrio lux* system in bioluminescent reporters -qRTPCR expression analysis of biosynthetic pathways in bacterial consortia (biofilms)
-Heterologous overexpression of genetically modified enzymes in various vectors.
-CRISPR knockouts of several critical enzymes in biopolymers synthesis in bacterial models and plants (Collaboration with Andreas Stürmer (T. Forschung), Austria.
-*In silico* analysis of biosynthetic pathways of PHBs in freshwater microalgae (Glyxon's proprietary research), implementation of the criteria ruled by NOM-059-SEMARNAT-2010 and NOM-172-SEMARNAT-2019
-Scale up production of Microbial cellulose for biosensors and materials applications (Glyxon's proprietary research)

Glyxon Biolabs is a Biotechnological DIYBio initiative promoting *open science projects*. Our global partners and collaborators encompass university labs,Fab labs, DIYBio organizations, independent researchers, biotechnological companies, and independent researchers organizations.

Some of our partners and collaborators:

**Ottawa BioScience** CANADA, **Cycese Mexico** MEXICO; **JOGL-CRI** FRANCE; **PipetteJockey/York University** CANADA; **ATG Belgium** BELGIUM; **Leibniz University** GERMANY; **WhiteBoxLabs** SWITZERLAND; **Gaudi Labs** SWITZERLAND; **Osaka University** JAPAN; **Open Bioeconomy Lab, Cambridge** UK, **Universidad de San Francisco de Quito** ECUADOR; **Applied Procrastination** NORWAY; **CNC Kitchen** GERMANY ; **Biocurious** USA; **Microchip Technologies**, Arizona USA,

**FELLOWSHIPS, GRANTS & AWARDS**

**2025, Twist Bioscience/MIT Media Lab David S. Kong's, Global Community Biotechnology Lab:** Sponsorship Award : $5,400 USD for Synthetic Biology projects (Shared with Ottawa-Open Bioscience Labs, Canada) Synthesis of novel solutions for biomaterials research and synthetic biology tools development. (USA, MEXICO, CANADA)

**2025, MIT Media Lab How to Grow Almost Anything (HTGAA) Course: David S. Kong (MIT), George Church (Harvard Univ), Joseph Jacobson (MIT). Global Committed Listener Fellowship** USA, Remote.

**2023, Experiment foundation,** "**How carbon sequestration is modulated by Mycorrhizal and soil bacteria symbionts in Abies trees?**" 4,400 USD Research grant for the study of mycorrhizal helping bacteria in soil carbon sequestration., USA, MEXICO

**2022, Internet of Production Alliance** February-December. Ongoing **Mapping the Digital and Additive manufacture capabilities in LatinAmerica** CANADA, USA, MEXICO, CHILE, BRAZIL **(Collaboration with Gregory Merritt, University of Victoria, B.C., Canada)**

**2021, MIT Media Lab Global Community Biofellows Program** October-2021 USA. **Global Leadership Development Program, Lecturer: Abel Cano, Kennedy School of Government, Harvard University.**

**OPEN COVID19 Initiative JOGL-Centre Recherches Interdisciplinaires (CRI)-AXA**, FRANCE
(R&D* grant, Project: Low Cost Chromatography Device: Proteopresso)

**2020 OPEN COVID19 Initiative JOGL-Centre Recherches Interdisciplinaires (CRI)-AXA**, FRANCE
(R&D* grant, Project: Open Source: Bioreactor)

**2019 Outstanding Paper Award 2019, Biophysical Society of Japan,** JAPAN

**2017 Complex Systems in Biology CONACYT PhD Fellowship,** MEXICO

**2012-2014 Department of Physics; School of Science and Engineering, Waseda University**; Tokyo, JAPAN
(Research Fellowship).

**2011-2012 GCOE Program to Support Young Researchers; Japan Society for the Promotion of Science** (JSPS), JAPAN.

**2010-2012** Protonic Nano machine Group, **Graduate School of Frontier Biosciences, Osaka University**, Osaka, JAPAN.

**Global Centres of Excellence Program (GCOE) "Biological Function Dynamic Systems".** (Research Fellow)

**2009-2010** Max Planck Gesellschaft, **Max Planck Institute for Terrestrial Microbiology**, Marburg, GERMANY,
(Postdoctoral fellowship).

**2008-2009 Graduate School of Biomedical Sciences UNAM**; PhD thesis publication support grant. Mexico City; MEXICO.

**2003-2008** CONACyT at IFC-UNAM; **Graduate School of Biomedical Sciences**; Doctoral Fellowship, Mexico City, MEXICO.

**2003** CONACyT / SEP * **IFC-UNAM**; Bs. Dissertation Fellowship. Mexico City, MEXICO.
R&D*: Research and Development

## CONFERENCES AND MEETINGS

**2024** Maker Faire CDMX, Talk: “What is synthetic biology”. National Center for Arts, MEXICO

**Nov 2023**, SMBE Satellite Meeting on Mechanisms of Cellular Evolution Symposium Arizona State University, Biodesign Institute: “Osmomodularion of MotA/B flagella motor rearrangement regulation in E.coli through visualization of member potential”. Supervisor of Undergrad students project at the Wadhwa lab (Poster). ARIZONA, USA

**2023** Maker Faire CDMX, Talk: “What is biohacking” National Center for Arts MEXICO

**2021** Global Community Biosummit 5.0, MIT Media lab , USA (On-line).

**Workshop:** “An Open Source Bioreactor and Chromatography system"
(Project developed in collaboration with Ottawa Bioscience and funded by JOGL, On-line event).
Global Community Biosummit 5.0, MIT Media lab , USA (On-line).
Responding to COVID-19 (Lighting talks, Panel moderator)

**2020** Global Community Biosummit 4.0, MIT Media lab , USA (On-line).
**Talk:** “Building a wave Bioreactor"
(Project developed in collaboration with Ottawa Bioscience and funded by JOGL, On-line event).

**2012** Annual Conference of Flagellum Biology. 9 to 11 March., Izu, JAPAN.
**Talk:** "Dynamic torque-speed flagellar motors disconnected mutants in MotA / B channels."
(Project developed in the Graduate School of Frontier Biosciences, Osaka University, JAPAN).

**2011** 49th Annual Meeting of the Biophysical Society of Japan. 16-18, Sep. 2011 University of Hyogo, Himeji, JAPAN.
**Poster** “Numerical simulation of 'unplugged' MotA/B flagellar stator channels in the engine."
UIPAB Biophysical Society International, International Biophysics 17ºCongress. October 30-November 3, Beijing CHINA.

**Poster:** "Dynamics of 'unplugged' MotA/B flagellar stator channels in the engine."
(Project developed in the Graduate School of Bio Sciences Frontera, Osaka University, JAPAN).

**2010** 48th Annual Meeting of the Biophysical Society of JAPAN. Sep. 20 -22
Kawauchi Campus, Sendai, Tohoku University, JAPAN. **Poster2009** BLAST-X (Locomotion Bacterial and Signal Transduction) Conference; Cuernavaca, MEXICO.

**Poster:** "Functional analysis of an extensive non-conserved region of FlgK (HAP1) of *Rhodobacter sphaeroides*".
(Project developed in the Department of Molecular Genetics IFC-UNAM, MEXICO).
36th Annual Conference of Biology of Myxobacteria, Boston MA, USA.

**Poster:** "Regulation of type IV pili assembly in *Myxoccocus xanthus*".
(Project developed at the Max Planck Institute for Terrestrial Microbiology, Marburg GERMANY).

**2006** XXIth IUBMB FAOBMB 11th International Congress and Congress of Biochemistry and Molecular Biology: "Life: Molecular Biological Diversity and Integration" Kyoto, JAPAN.

**Poster:** "Functional characterisation of FlgK (HAP1), structural component of the flagellum of *Rhodobacter sphaeroides*"
(Project developed in the Department of Molecular Genetics IFC-UNAM, Mexico City MEXICO).
XXVIth National Congress of the Mexican Society of Biochemistry, Guanajuato, MEXICO.

**Poster:** "Study of the interface hook-filament of *Rhodobacter sphaeroides*".
(Project developed in the Department of Molecular Genetics IFC-UNAM, MEXICO / University of Osaka, JAPAN).

**2004** XXVth National Congress of the Mexican Society of Biochemistry, Ixtapa-Zihuatanejo, Guerrero. MEXICO.
**Poster:** "Study of the interface hook-filament interface of *Rhodobacter sphaeroides*"
(Project developed in the Department of Molecular Genetics IFC-UNAM, MEXICO).

**2002** XXIVth National Congress of the Mexican Society of Biochemistry, Puerto Vallarta, Jalisco, MEXICO.
**Poster:** "Genetic and Molecular Study of HAP1 protein structural component of the flagellum of *Rhodobacter sphaeroides*” (Project developed in the Department of Molecular Genetics IFC-UNAM, MEXICO).

**2000** Society for Neuroscience (Society for Neuroscience) 30th Annual Conference. New Orleans LA. USA.

## SCIENTIFIC PUBLICATIONS

**2023** Gomez-Hinostroza ES, Gurdo N, Alvan Vargas MVG, Nikel PI, Guazzaroni M-E, Guaman LP, ***Castillo Cornejo DJ***, Platero R and Barba-Ostria C (2023) Current landscape and future directions of synthetic biology in South America. Front. Bioeng. Biotechnol. 11:1069628. doi: 10.3389/fbioe.2023.1069628

**2013** ***David J. Castillo****, Shuichi Nakamura, Yusuke V. Morimoto, Yong-Suk Che, Nobunori Kami-ike, Seishi Kudo, Tohru Minamino and Keiichi Namba.* "The C-terminal periplasmic domain of MotB coordinates the number of stators in the bacterial flagellar motor in response to changes in external load". Biophysics*. Vol. 9, pp. 173–181 (2013) doi: 10.2142/biophysics. 9.173 (* The official on-line journal of The Biophysical Society of Japan BSJ).

**2009** ***Castillo DJ****, Ballado T, Camarena L, Dreyfus G*. "Functional analysis of a large non-conserved region of FlgK (HAP1) from *Rhodobacter sphaeroides*". Antonie Van Leeuwenhoek. 2009 Jan;95(1):77-90.

## REFERENCES

**Prof. Keiichi Namba, Ph.D.**
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Suita, Osaka 565-0871, **Japan**
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**Prof. Carlos Barba-Ostria Ph.D.**
Escuela de Medicina, Universidad San Francisco de Quito,
Quito, **Ecuador**
cbarbao@usfq.edu.com

*Language Skills:* Fluent in English, Spanish, Intermediate Skills in French and Japanese, Basic German

*Lab Skills:* PCR, qPCR, Electrophoresis of nucleic acids. Souther/Northern blot hybridization. Design and cloning of recombinant proteins by traditional restriction digestion cloning, GoldenBraid, SLiCe and Gibson assembly. SDS-PAGE/Native gels for Protein electrophoresis, IPh-2D electrophoresis, Westernblot, Preparation of antibodies (mice and rabbit). Protoplasm isolation, liposomes preparation, Cell fractionation of cellular membranes and organelles isolation by differential centrifugation and ultra centrifugation. Design and construction of microfluidics by hard and soft elastomers (Acrylate and PDMS). Bio/ chemical functionalization of surfaces (glass, plastics, metal). Transfection of plants with bacterial vectors (*Agrobacterium* infiltration). Single particle analysis and cell motility analysis using high speed nanophotometry. Microscopy: Bright field, Darkfield, TIRF, confocal and deconvolution miscoscopy. Basic operation of TEM. Plant tissue propagation in liquid and semisolid media. Stereotactic neurosurgery of rats, (neurochemical ablation trough excitotoxin injection , cannula and electrodes implantation)

*Computer:* Proficient in Office, AutoCAD, Eagle CAD, Fusion 360, Adobe Illustrator, Phyton, Arduino/RaspberryPi, COMSOL-Multiphysics. Igor and Matlab. PCB design: Circuit Maker, Fritzing, KiCAD. Microfluidics design: Flui'device, Flui3d. Digital microfludics: Opendrop. Additive manufacture: Tinkercasd, Cura, Simplify 3D.

*Interests:* Analog photography, Electronic music production, Painting, Ceramics, Hiking, and Wildlife macro photography. Sometimes make silly robots.